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Probing the solar coronal magnetic field with physics-informed neural networks

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While the photospheric magnetic field of our Sun is routinely measured, its extent into the upper atmosphere is typically not accessible by direct observations. Here we present an approach for coronal magnetic-field extrapolation, using a neural network that integrates observational data and the physical force-free magnetic-field model. Our method flexibly finds a trade-off between the observation and force-free magnetic-field assumption, improving the understanding of the connection between the observation and the underlying physics. We utilize meta-learning concepts to simulate the evolution of active region NOAA 11158. Our simulation of 5 days of observations at full cadence (12 minutes) requires less than 12 hours of total computation time, allowing for real-time force-free magnetic-field extrapolations. The application to an analytical magnetic-field solution, a systematic analysis of the time evolution of free magnetic energy and magnetic helicity in the coronal volume, as well as comparison with extreme-ultraviolet observations, demonstrates the validity of our approach. The obtained temporal and spatial depletion of free magnetic energy unambiguously relates to the observed flare activity.

Extrapolation and simulation methods are used to provide an estimate of the three-dimensional distribution of the coronal magnetic field[1], which is essential to understand the genesis and initiation of solar eruptions and to predict the occurrence of high-energy events from our Sun[2]. The most realistic approximations to the magnetized plasma under coronal equilibrium conditions (where gas pressure and other forces are negligible), are nonlinear force-free (NLFF) magnetic-field models[3]. The computation of a force-free magnetic field, **B**, inside a simulation volume, requires the numerical solution of the following system of differential equations

$$\mathbf{J} \times \mathbf{B} = 0, \tag{1}$$

where **J** is the electric current density, and

$$\nabla \cdot \mathbf{B} = 0 \tag{2}$$

subject to values specified through direct observations on the volume's bounding surface. In practice, data at sufficiently high spatial and temporal resolution are obtained from measurements of the magnetic-field vector **B** at photospheric heights only (at $z = 0$; bottom boundary of the simulation). The data are known to be inconsistent with the force-free assumption (for example, ref. 4). For compensation, these data are usually specifically processed before extrapolation of the coronal field (for example, refs. 5,6), and/or the NLFF solution may deviate from the supplied boundary data at $z = 0$ (for example, refs. 7–10). The boundary values on the remaining (side and top) boundaries of a usually cubic computational volume are most commonly pre-set by a potential-field solution, which can be computed from the observed data at $z = 0$. These adjustments generally improve the force-free quality of the NLFF solution for a given set of boundary conditions[5], yet do they not guarantee to achieve a force-free solution[11]. Moreover, the potential field prescribed at the boundaries certainly represents only an approximation to the true coronal field (compare, 'Comparison with analytical NLFF fields' section).

Deep learning applications are frequently used to learn a general task by using vast amounts of labelled data (supervised training). Physics-informed neural networks[12] represent a different class that can

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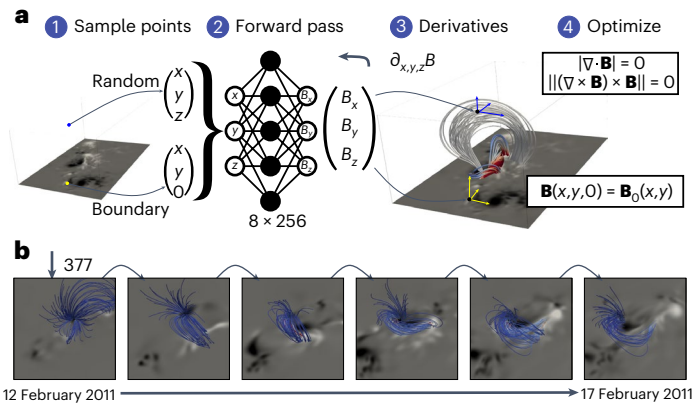

Fig. 1 | Overview of the proposed method for force-free magnetic-field extrapolation. a, Our model training is iteratively performed in four steps. We start by sampling points from the boundary condition ($x, y, z = 0$) and randomly from the volume (x, y, z). Each point is passed through our fully connected SIREN network[34], which consists of 8 hidden layers with 256 neurons each. The output of the network are the corresponding magnetic-field vectors at the sampled points (B_x, B_y, B_z). For each vector **B**, we compute the derivatives with respect to the input coordinates, using automatic differentiation. The derivatives are used to construct the differential equations, which serve as loss function. We optimize our model to satisfy the divergence-free and force-free equation in the entire volume, and to match the input field at the boundary. With the iterative application of these steps, we find the magnetic-field extrapolation. **b**, For the simulation of time series, we perform a full extrapolation of the first frame. The consecutive frames use the previous results as the starting condition. This allows to efficiently adapt changes and rapidly converge to the next solution. We illustrate the evolution of active region NOAA 11158 over 5 days, and trace the field lines of the negative polarity sunspot. The series shows the evolution of a potential-like field into a strong current-carrying flux rope and its decay back. The colour of the field line traces shows the magnitude of electric current density, where strong and weak currents are indicated in red and blue, respectively.

solve nonlinear partial differential equations through a data-driven approach, uniquely integrating physical models and data in the framework of deep learning. This method has shown the ability to find meaningful solutions even when the problem is not perfectly posed[13], and has been successfully applied for various problem settings (for example, refs. 14–16).

We aim at finding a solution to the boundary-value problem by including the NLFF physical equations in a neural network. Here the neural network acts as function that maps each coordinate point in the simulation volume (x, y, z) to the corresponding magnetic-field vector (B_x, B_y, B_z). In this way, the neural network serves as a mesh-free representation of the simulation volume. At the simulation boundary $z = 0$, we match the values of our neural network with the grid points of the observation $\mathbf{B}_0(x, y)$, such that we obtain a neural representation that is a smooth approximation of the boundary magnetic field. For the entire volume, we optimize our model to satisfy the force-free assumption as given by the system of differential equations (equations (1) and (2)), which determines the coronal extrapolation of the observed photospheric magnetogram. For the construction of the differential equations, the derivative of **B** with respect to the input coordinates is computed from the neural representation using automatic differentiation (Methods). In addition to the force-free constraint, we require that the magnetic field matches a potential field at the side and top boundary (compare, ref. 7), which prevents artificial fields in weak-field regimes close to the top boundary. For the training, we iteratively sample a set of 10^4 boundary points and 2×10^4 random points from the entire volume and optimize our model to minimize the deviation from the boundary condition and the physical model simultaneously (Fig. 1). By sequentially applying these update steps, a three-dimensional extrapolation of the photospheric magnetic field is found.

The assumption of a force-free magnetic field does not hold for photospheric heights[4,17] and observations are subject to noise, instrumental characteristics and the Stokes inversion that is being used[18]. Therefore, both the data and the underlying physical model of our optimization problem are imperfect. Physics-informed neural networks have the advantage to facilitate an intrinsic adjustment of the boundary condition and to allow for deviations from the physical model, where the solution is determined by the weighting between both components (Methods). For our application, this enables us to omit any pre-processing steps of the input data. Instead, we find dynamically an approximation of the boundary condition that is in agreement with the force-free assumption, and vice versa deviate from the physical model where the boundary field is enforced.

Results

To demonstrate the applicability and soundness of our approach, we compare the results from the network with an analytical magnetic-field solution and with observational data. We start by assessing the ability to find a solution to the force-free problem under idealized conditions ('Comparison with analytical NLFF fields'). In 'Trade-off between observation and physical model', we address the intrinsic property of our method to find solutions that account for the imperfections of the data and the physical model. In 'Simulation of time series', we provide a systematic comparison of the time evolution of magnetic energy and magnetic helicity derived from the model field. We use extreme-ultraviolet (EUV) observations to relate to flare events and to compare with the magnetic topology of our simulation.

Comparison with analytical NLFF fields

We apply our method to the 'Low and Lou' model[19], which is analytical except for the numerical solution of an ordinary differential equation in spherical coordinates with eigenvalues m and n (therefore, representing a semi-analytical test case). The obtained fields are axisymmetric, but by rotating the symmetry axis by a rotation angle Φ and placing the singularity at a depth l below the photosphere, we can mimic coronal fields that are useful for assessing the performance of NLFF models for the case of large-scale smooth currents distributed in the entire test volume[3,20,21]. In ref. 20, six different state-of-the-art NLFF methods for the three-dimensional coronal field extrapolation (including optimization, Grad–Rubin, Green's function and magnetofrictional) have been systematically compared in blind tests on such deigned test cases. Here we use Case I from ref. 20, and the test field from ref. 7, which we define as C1 and C2, respectively. C1 uses $n = 1$, $m = 1$, $l = 0.3$ and $\Phi = \pi/4$, in a volume spanning 64^3 grid points, and where we provide the solution on all 6 boundaries to our method. For C2, we set $n = 1$, $m = 1$, $l = 0.3$, $\Phi = 0.15\pi$ and define a computational volume spanning $80 \times 80 \times 72$ ($x \times y \times z$). We provide only the bottom boundary of the such designed test field as input to our method and evaluate the central 64^3 cube (compare, ref. 7).

This ideal solution allows us to perform a quantitative comparison, for which we use the metrics introduced in ref. 20 (see 'Analytic field solution' in Methods). We consider four metrics that compare local characteristics (the vector correlation (C_{vec}), the angular difference (C_{CS}), the complement of the normalized vector error ($E'_{\mathrm{n}} = 1 - E_{\mathrm{n}}$) and the complement of the mean vector error ($E'_{\mathrm{m}} = 1 - E_{\mathrm{m}}$)), where larger values indicate a better agreement, and where a maximum of unity means that the semi-analytical test field and our numerical solution are identical. The global similarity to the reference field is estimated by the ratio of the total magnetic energies (ϵ). In Table 1, we summarize the results. For C1, our method achieves a solution almost identical to the test field (Supplementary Fig. 1). By using only the bottom boundary as an input (C2), the NLFF extrapolation solution shows larger deviations from the ideal solution, highlighting the limitations

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Table 1 | Quantitative comparison of our method to the semi-analytic magnetic-field solutions from the Low and Lou model

Test case	C_{vec}	C_{cs}	E'_n	E'_m	ϵ	$\sigma_J \times 10^2$
C1	1.00	1.00	1.00	0.99	1.00	1.99
C2	1.00	0.93	0.84	0.68	1.07	3.51

For C1, we use all boundaries as input, whereas for C2, we use only the bottom boundary, where the remaining boundaries are approximated by a potential-field solution. We compare the vector correlation (C_{vec}), angular difference (C_{CS}), the complement of the normalized vector error ($E'_n = 1 - E_n$), the complement of the mean vector error ($E'_m = 1 - E_m$), relative total magnetic energy (ϵ), and the current-weighted average of the sine of the angle between the magnetic field and electric current density (σ_J).

introduced by using a potential-field approximation for the side and top boundaries. We note that our approach gives the highest performance compared with the six methods for NLFF extrapolations that were systematically tested on the same analytical solution (C1) in ref. 20. We note that these methods have been subject to improvement (for example, refs. 7,9), but similar comparative studies of this test case have not been performed. For C2, we find that our method performs similarly successful as existing state-of-the-art NLFF methods.

Trade-off between observation and physical model

We apply our method to the active region NOAA 11158 (SHARP 377). This region was observed from its emergence (12 February 2011), over more than 5 days by the Helioseismic and Magnetic Imager (HMI)[22] onboard the Solar Dynamics Observatory (SDO)[23]. The active region developed into a complex quadrupolar sunspot formation that led to multiple solar flare eruptions, including an X2.2-class flare (SOL2011-02-15T01:44; peak-time, 15 February 2011 at 01:56 UT), originating from the main polarity inversion line along which a sigmoidal filament was observed. An M2.2-class flare (SOL2011-02-14T17:20; peak-time, 14 February 2011 at 17:26 UT) originated from the eastern part of the active region (for a detailed analysis, see ref. 24).

In Fig. 2, we study the trade-off between the physical model and observed data, where we define the weighting parameters for the divergence-free, force-free and boundary conditions, as λ_{div}, λ_{ff} and λ_B, respectively. In Fig. 2b, we compare nine different parameter settings, with varying $\lambda_{div/ff}$, full and factor-of-two binned resolution of the input magnetogram, and different conditions for the side and top boundary. The plots on the left show the divergence-free (upper) and force-free (lower) error over the difference to the observation data (B_{diff}; Methods). The plots on the right show the normalized metrics, $|\nabla \cdot \mathbf{B}|/|B|$ and the average angle between the currents and magnetic field (θ_j; Methods). Therefore for all plots, points towards the left are in better agreement with the observed photospheric magnetic field, whereas points towards the bottom are in better agreement with the physical model. All metrics show that the ideal point in the lower left is not attainable for the given boundary-value problem. The simulations are distributed along a supposed trade-off line, which determines the attainable solutions. This necessary trade-off is known from existing state-of-the-art NLFF modelling approaches (for example, ref. 25). Enforcing the constraints of the physical model leads to smoother solutions for the magnetic field with diminished signal on smaller spatial scales, whereas reducing the physical constraints gives a better approximation of the observation (Fig. 2a).

The use of the full-resolution observational data (0.36 Mm per pixel) shows that the higher spatial information only results in a shift along the trade-off line. This suggests that the degree of force freeness does not profit from the higher spatial resolution, and that the parameter choice dominates the result (compare, ref. 25). It is worth noting that using stronger binning could lead to substantially different results (as demonstrated in ref. 11). The main rationale for using binning is

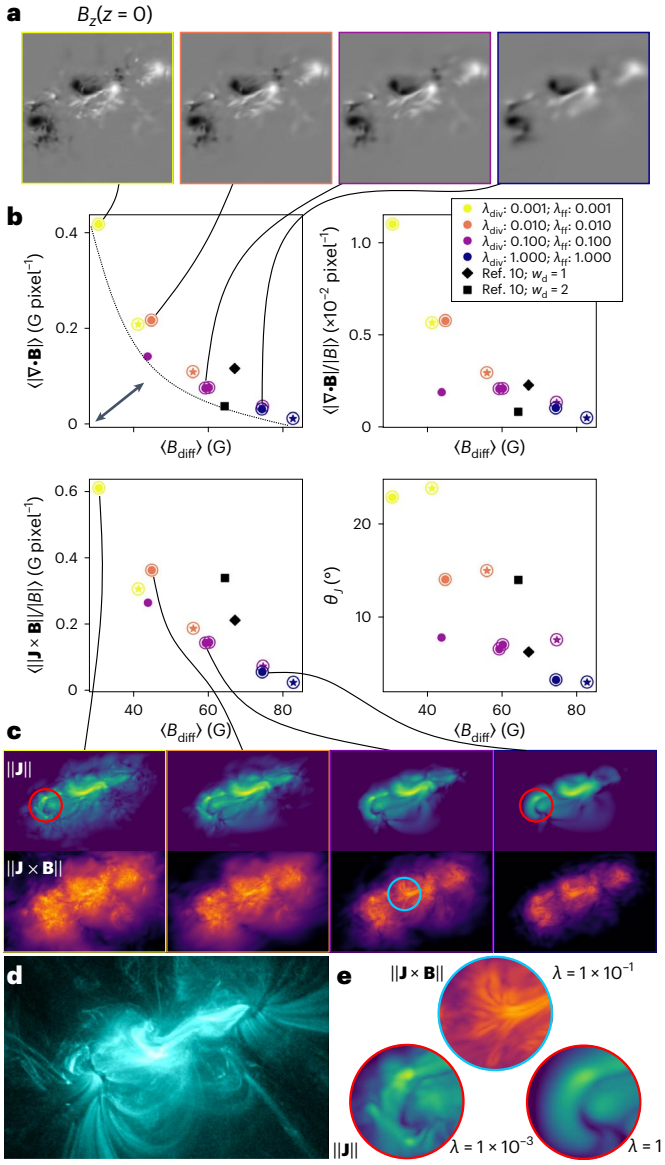

Fig. 2 | Evaluation of different parameter settings and trade-off between measurement and physical model, using the vector magnetogram from 15 February 2011 at 00:00 UT. a, Increasing the weighting of the physical model $\lambda_{div/ff}$ decreases the similarity to the observed magnetic field. The adjustment of the boundary is model intrinsic to satisfy the force-free assumption. The series of four images shows the change of the bottom boundary with increasing weighting of the force-free constraint. **b**, Metrics on the *y* axis represent deviations from the physical model. The *x* axis gives the deviation from the observation. The plots show the divergence-free (upper left), normalized divergence-free (upper right) and force-free (lower left) error, as well as the average angle between the currents and the magnetic field (lower right). The plain dot symbol refers to an open boundary simulation, whereas the other settings (encircled symbols) use a potential-field upper boundary. Stars refer to simulations that use the full spatial resolution (0.36 Mm per pixel), dots use data binned by a factor of two and black symbols refer to a classical state-of-the-art NLFF method for comparison, where w_d refers to the weighting parameter of the divergence that was used for the extrapolation (Supplementary Section B). The dashed line illustrates the supposed trade-off. **c**, Maps of integrated current density (top) and integrated horizontal component of $||\mathbf{J}\times\mathbf{B}||$ (bottom). The different parameter settings result in a similar magnetic topology. **d**, Reference EUV observations of the SDO/AIA 131 Å channel. **e**, Comparison of cut-outs from **b**, where colours indicate the corresponding regions. Top: deviations from the physical model occur at spatially confined regions and appear highly structured. Bottom: small spatial structures are enhanced when we reduce the constraints of the physical model.

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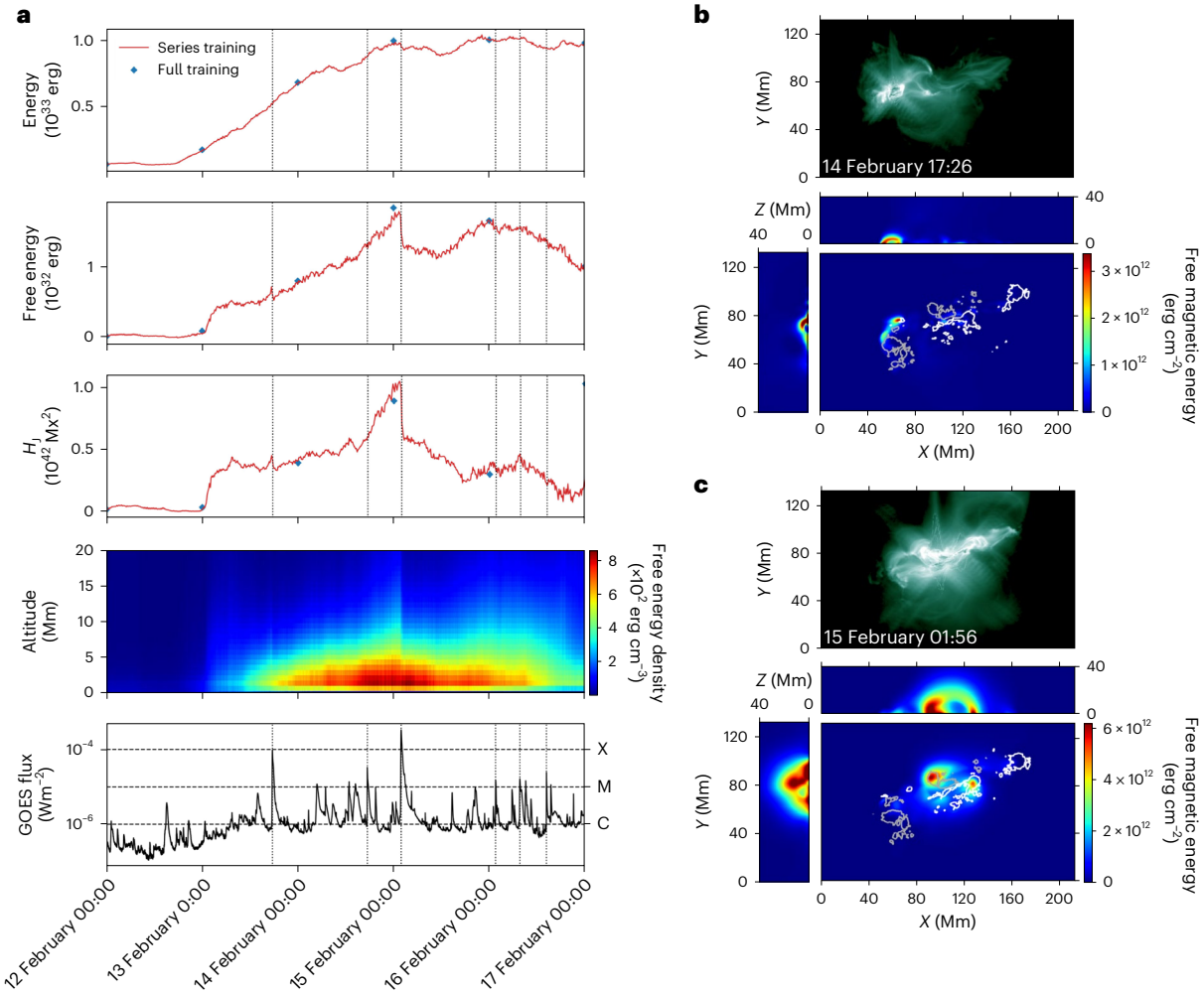

Fig. 3 | Simulated series of active region NOAA 11158 and comparison of extracted parameters with observations. a, The integrated total magnetic energy, free magnetic energy, magnetic helicity (H_J), free-energy density distribution in height and the GOES 1.0–8.0 Å flux. Blue symbols indicate simulations from scratch, that lead to similar results as the series training (red line). The signature of the X-class flare is clearly visible as a sharp jump in free magnetic energy and helicity. Flares ≥M1 are indicated by vertical lines. The associated contraction of loops can be seen in the height distribution of free energy. The majority of free magnetic energy is stored low in the model corona. **b**,**c**, Time-integrated EUV observations (top) and the corresponding maps of released free magnetic energy during the flare event (bottom). Contour lines at ±1,000 G are overlayed in grey (negative) and white (positive). **b**, M-class flare on 14 February 2011 between 17:20 UT and 17:32 UT. The depleted free magnetic energy is spatially associated with the observed flare. A weak energy release along the central polarity inversion line can be seen in the EUV observations, which is also present in our simulation. **c**, X-class flare on 15 February 2011 between 01:44 UT and 02:06 UT. The event is associated to a strong energy release along the central polarity inversion line. The loop-like structure in the x–z plane originates from the contraction of the magnetic field during the flare event. A video of the full series is provided in Supplementary Video 1.

usually the computing time. In our update scheme, however, the choice of binning does not directly affect computing time.

The usage of a potential field as a boundary condition at the side and top boundaries of the model volume, as commonly used in existing NLFF methods, is not strictly required for our method. In Fig. 2b, the simulation with an open upper boundary leads to a supposed improvement orthogonal to the trade-off line (dashed line). We found that for regions of weak fields, this constraint is necessary to prevent artificial fields close to the upper boundary. Therefore, an additional constraint of the side and top boundary is required to provide valid solutions. The improvement orthogonal to the trade-off line suggests that the potential-field approximation causes a further trade-off with respect to the observed photospheric magnetic field.

Independent of the tested choices of the weighting parameters (λ_{div}, λ_{ff} and λ_{B}), all simulations reveal a similar magnetic topology as illustrated by the maps of vertically integrated current density (Fig. 2c). The main difference occurs at smaller scales, where the field appears more structured when the constraints of the physical model are loosened (Fig. 2e). The strong current-carrying magnetic fields close to the polarity inversion lines can be also observed in the corresponding EUV observation in Fig. 2d.

In Fig. 2c, we show the residual of the horizontal $\|\mathbf{J}\times\mathbf{B}\|$ term, which we associate with regions where the force-free assumption can not be satisfied by our model. Relaxing the physical constraints leads to an overall increase of the error, but also to a separation into smaller regions. The blue cut-out in Fig. 2e shows the central flux rope, where the largest deviations occur at the intersection of magnetically separated regions (that is, flux rope and overlying field).

The optimization allows for a certain variability between individual runs. In Fig. 2b, we compare two runs with $\lambda_{\mathrm{div/ff}} = 0.1$, which demonstrates that our method converges to similar solutions, and that significant differences only originate from the selected weighting.

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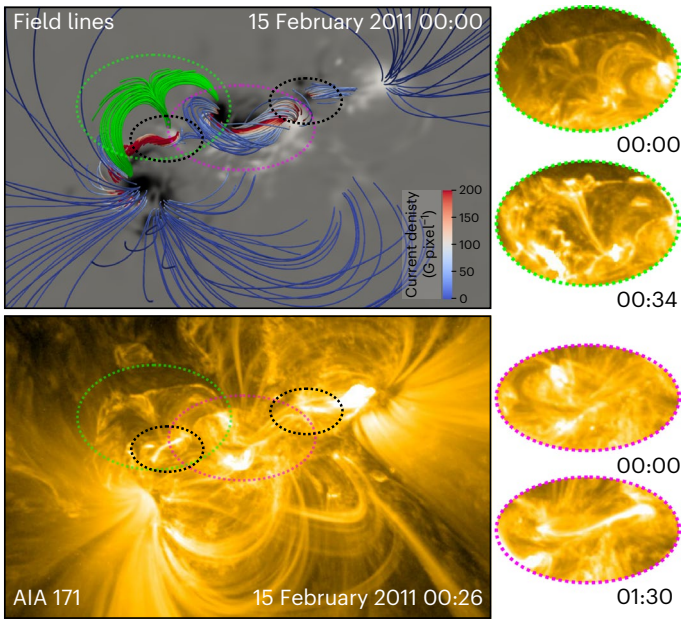

Fig. 4 | Simulated magnetic topology of active region NOAA 11158 and comparison with observations in the EUV. Top: magnetic-field line traces of the central components of the active region. The colours of the field lines indicate the current density. The central flux rope is encircled in pink, two smaller flux ropes in black and the eruptive structure in green. Bottom: the corresponding regions are also indicated in the SDO/AIA 171 Å observation. We show the primary regions of interest at different time steps during the eruption (right), where the identified magnetic structures become observable in the EUV.

Simulation of time series

To obtain a robust understanding of the temporal evolution of the solar corona before eruptive events, the computation of time sequences of NLFF models is required. Traditional state-of-the-art methods do so by solving the NLFF boundary-value problem individually for each time step. Here we assume that the model parameters of our neural network are close to temporally adjacent extrapolation results. We start with a full simulation of the initial frame of the series, while the following time steps are constructed by using the previous simulation as initial model parameters (Fig. 1b). This enables a convergence to the next solution that is typically faster than the instrumental cadence of 12 min (see 'Series training' in Methods). Figure 3a demonstrates that this training method (solid line) leads to similar results as performing a training run from scratch (blue symbols).

We apply our method (using $\lambda_{\mathrm{div/ff}} = 0.1$) to 601 consecutive magnetograms at a time cadence of 12 min, covering the time period from 12 February 2011 at 00:00 UT to 17 February 2011 at 00:00 UT. In Fig. 3a, we show the corresponding time profiles of total magnetic energy, free magnetic energy and magnetic helicity of the current-carrying field (H_J; Methods), together with the Geostationary Operational Environmental Satellite Network (GOES) soft X-ray flux (bottom) as a reference for the occurrence of large flares. Throughout the evolution of the active region, the total energy gradually increases. In the time profile of the free magnetic energy, and even more pronounced in that of the helicity of the current-carrying field, clear responses to the occurrence of solar flares are identified. The free magnetic energy density is in agreement with other methods that report that free energy is stored low in the solar corona[24]. The sharp jump in the free magnetic energy density (15 February at 02:00) is consistent with a flare-related implosion, and can be observed as contraction of the coronal loops (Supplementary Video 1). For our computation environment with four NVIDIA V100 graphics processing units (GPUs), a simulation from scratch requires ~45 min and the series extrapolation requires ~1 min per frame, with a total run time for the full series of 12 h (including data processing).

In Fig. 3b,c, we show spatial maps of depleted free magnetic energy during two flare events, where we integrate along the *x*-, *y*- and *z* axes, respectively. Here we only consider energy losses, which allow for an easier identification of the flares as from the integrated quantities in Fig. 3a. The top row shows the integrated EUV emission over the duration of the flare, which emphasize the regions of energy release ('Time-integrated maps' in Methods). Figure 3c shows the X-class flare that occurred close to the central polarity inversion line. Figure 3b shows the M-class flare that originated from the smaller polarity inversion line in the eastern part of the active region. For this event, the flare energy release was not visible in the spatially integrated free-energy curves. For both flares, the free-energy depletion aligns with the position of the flare. We note that also regions revealing a small energy release (light blue) can be associated with enhanced EUV emission. For the M-class flare an energy release can be noted at the centre, aligning with the minor brightness enhancement in EUV (that is, central flux rope on 14 February at 17:26).

In Fig. 4, we compare magnetic-field traces of our simulation with observations from SDO/Atmospheric Imaging Assembly (AIA). The central flux rope becomes clearly visible in 171 Å, minutes before the major flare event (pink circle). Similarly, the magnetic structure, indicated by a green circle, appears very faint in the EUV emission and becomes prominent close to the eruption (15 February at 00:34), where we note a good agreement with our modelled magnetic field. Our model suggests that this structure built up over hours before the eruption. We provide a more extensive comparison in Supplementary Section B. In Supplementary Video 1, we show the evolution of the free magnetic energy, current density and running difference of free magnetic energy, for the entire series.

According to our model, the majority of free-energy is stored a few Mm above the polarity inversion line (compare, Supplementary Video 1), where also a large fraction of magnetic energy is depleted during the flare event (Fig. 3b). The change in free energy during the X-class flare above ~5 Mm is probably related to the contraction of the magnetic field (magnetic implosion)[26]. The M-class flare shows an energy release only from heights below ~5 Mm. Both events show the same characteristic energy release along the flux rope (compare, Fig. 3c).

We note that the training from scratch and the series training give different results for the last time step in the magnetic helicity (H_J) series. For this period, strong magnetic fields terminate at the side boundaries, suggesting that the corresponding boundary condition is reflected by the unexpected variation in helicity. We assume that the boundary region is less constrained and evolves differently in the series training, which causes the difference in magnetic helicity.

Discussion

The comparison between model-deduced flare-related changes and coronal (EUV) observations suggests a suitable approximation of the coronal magnetic field. As evaluated by standard metrics, our method achieves state-of-the-art NLFF magnetic-field extrapolations (Table 1 and Fig. 2b). The further application to observational data from 66 different solar active regions shows that our method delivers high-quality solutions, independent of the magnetic configuration and area of the considered active region (Supplementary Section D), and provides the ability to achieve magnetic-field extrapolations in quasi real time (see 'Series training' in Methods). Extracted parameters from vector magnetograms are frequently used for solar flare prediction[27] and parameters from three-dimensional vector magnetograms have been demonstrated to provide reliable precursors for large flares[28]. With the fast extrapolation of time series, our method can provide the datasets required to extend these efforts towards data-driven methods, and automatic prediction of solar flares.

In ref. 24, the active region NOAA 11158 was analysed in detail, using a series of magnetic-field extrapolations, similar to our estimation. From Fig. 3, we report a similar rapid free-magnetic-energy injection, followed by a gradual energy build-up, and a step-wise energy release during the X-class flare. We also find that the free magnetic energy is stored low in the solar atmosphere. Differences are primarily related to the estimated free magnetic energy, where we obtain a peak value of 1.8×10^{32} erg, as opposed to the 2.5×10^{32} erg reported in ref. 24. The energy drop during the X-class flare is 0.5×10^{32} erg, which corresponds to a relative free-magnetic-energy depletion of 20% over 1 h, and is larger than the 0.3×10^{32} erg (14%) reported in ref. 24. We further note that our quality metrics for divergence freeness and force freeness are in the same range as the well-tested and widely used optimization-based extrapolation method in ref. 7, and that the evaluated E_{div}/E is below the tolerance limit of ~0.1 (with E_{div}/E quantifying the fraction of the total magnetic energy E which is related to the non-zero divergence case; for more details, see refs. 29,30 and Supplementary Section B).

A further notable advantage of our method is the file size of the resulting extrapolations, which requires about 2 MB. This corresponds to a compression by a factor of 60, compared with storing the mesh representation, which allows for easy distribution. The neural representation stores the full field information in the optimizable parameters, which provides a more memory efficient storage than a grid-based representation. Furthermore, arbitrary regions can be subsampled, and smooth derivatives can be computed, by sampling specific coordinates from the neural network.

As shown by the estimated trade-off line (Fig. 2b), the primary restrain of our method is the incompatibility between the underlying physical model and the observation. An extension of the physical model could be achieved by, for instance, accounting for the gradient of the gas pressure (non-force-free magnetic field). This would require further observational constraints to provide physically meaningful results, such as traced coronal loops[31] or differential emission measure estimates. Flares are very dynamic processes, and therefore even higher time cadences would be needed than are currently available (12 min). Similar to for the spatial extrapolation, the magnetic field of the intermediate time steps could be interpolated by extending towards a time-dependent model. Another natural next step is the application to magneto-hydrodynamic equations.

Methods

We employ a physics-informed neural network that uses the force-free assumption to simulate the stratification of the active region's magnetic field, based on the observed photospheric vector magnetogram. Here the neural network acts as a neural representation of the magnetic field at each point in the spatial simulation volume.

We consider the neural network as function mapping from the coordinate domain to the magnetic-field vector ($\mathrm{NN} : \mathbb{R}^3 \mapsto \mathbb{R}^3$; with NN being the neural network). The model training is performed for a single active region, by minimizing the deviation from the observed magnetic-field vector at the bottom boundary, the force-free loss (normalized by $\mathbf{B}$)

$$L_{\mathrm{ff}} = \frac{\| (\nabla \times \mathbf{B}) \times \mathbf{B} \|^2}{\| \mathbf{B} \|^2 + \epsilon}, \tag{3}$$

and the divergence-free loss

$$L_{\mathrm{div}} = |\nabla \cdot \mathbf{B}|^2, \tag{4}$$

in the entire volume (compare, equations (1) and (2)). Here, $\mathbf{B}$ refers to the neural network representation of the magnetic-field vector at a given point (that is, $\mathbf{B} = \mathrm{NN}(x, y, z)$) and $\epsilon = 1 \times 10^{-7}$ is added for numerical stability. We compute the derivatives of the magnetic field with respect to the input variables ($\frac{\partial B_x}{\partial x}, \frac{\partial B_x}{\partial y}, \frac{\partial B_x}{\partial z}, \ldots$). As our neural network corresponds to a fully differentiable function $\mathbf{B}$, the derivatives can be automatically computed using PyTorch. In contrast to grid-based approaches (for example, finite differences), this allows us to directly compute smooth derivatives at arbitrary positions without sampling the surrounding grid cells.

The driver of the simulation is the boundary condition $\mathbf{B}(x, y, 0) = \mathbf{B}_0(x, y)$, where $\mathbf{B}_0$ refers to the observation. We optimize for this condition by sampling coordinates from the $z = 0$ layer and minimizing the component-wise distance to the corresponding observation from the vector magnetogram

$$\mathbf{B}_{\mathrm{diff},0} = \mathrm{abs}(\mathbf{B} - \mathbf{B}_0). \tag{5}$$

In addition, we account for uncertainties in the measurement by subtracting the error map $\mathbf{B}_{\mathrm{error}}$ and clipping negative values (that is, optimizing only points that exceed the error threshold)

$$\mathbf{B}_{\mathrm{diff,clipped}} = \max\{\mathbf{B}_{\mathrm{diff},0} - \mathbf{B}_{\mathrm{error}}, 0\}. \tag{6}$$

For the loss of the boundary magnetic field, we compute the vector norm of the clipped difference vector

$$L_{B0} = \| \mathbf{B}_{\mathrm{diff,clipped}} \|^2. \tag{7}$$

In this formulation, the side and top boundaries of the cubic simulation domain are not explicitly bound. We assume that the magnetic field approaches a potential-field solution at those boundaries. For this, we compute the potential-field solution using the Green's function approach as proposed in ref. 32. This condition is imposed via the same loss terms as for the bottom boundary (equation (7)), where we assume no error.

We use Spaceweather HMI Active Region Patches (SHARPs)[33] that are remapped to a cylindrical equal area (that is, equidistant coordinate system). The three vector components ($B_{\mathrm{p}}, -B_{\mathrm{t}}, B_{\mathrm{r}}$) serve as the bottom-boundary condition, where B_{p}, B_{t} and B_{r} refer to the phi (westward), theta (southward) and radial (out of photosphere) component of the magnetic field, respectively. Their corresponding error estimates are used in the loss function as B_{errror}. Per default, we bin the data by a factor of two in both spatial dimensions. We normalize the magnetic-field strength to 2,500 G and the spatial coordinate dimensions to 160 pixels (320 for the original resolution). Values >1 are considered as input and output. For all our evaluations, we transform the magnetic extrapolation back to gauss units and use the mesh representation by sampling all coordinate points in the simulation volume.

For observations of NOAA 11158 (SHARP 377), we crop the active region patches to x [66:658] and y [9:377] pixels (before the binning), in correspondence with the optimization-based[3] (Supplementary Section A) NLFF modelling used for comparison (for a similar model time series see, for example, ref. 30).

For our model, we adapt the Sinusoidal Representation Networks (SIREN) architecture from ref. 34, where we use sine activation functions throughout our network, which are ideal for representing derivatives. The sine activation showed a slightly better performance than the tanh activation functions that were originally proposed for physics-informed neural networks[12]. Our model takes the 3 spatial coordinates as input and maps them over 8 fully connected layers, with 256 channels each, to the 3 output components of the magnetic-field vector. We omit the activation function for the output. Our evaluation suggests that a larger number of free parameters does not lead to a significant performance increase of our method (Supplementary Fig. 2).

While a positional encoding in terms of Fourier features[35] leads to a faster fitting of the high-frequency features of the $z = 0$ vector magnetogram, we found that it is not beneficial for the extrapolation of the magnetic field that is solely determined by the physical equations.

The training is performed iteratively, where we randomly sample 30,000 points from the simulation volume (batch size). Ten thousand points are taken from grid positions of the boundary coordinates and are used to compute L_{B0} (equation (7)). The remaining 20,000 points are randomly and mesh-free sampled from the spatial volume ($x = [0, x_{\max}]$, $y = [0, y_{\max}]$, $z = [0, 159]$ pixels; according to the bin-2 SHARP pixel size of 0.72 Mm) according to a uniform distribution. The L_{ff} and L_{div} loss is computed from all sampled points (equations (3) and (4)). We optimize the neural network for the combined loss

$$L = \lambda_{\mathrm{ff}} L_{\mathrm{ff}} + \lambda_{\mathrm{div}} L_{\mathrm{div}} + \lambda_{B0} L_{B0}, \tag{8}$$

where the λ parameters refer to the weighting of the individual losses.

The use of the original spatial resolution is straightforward, by using the full resolution of the boundary grid and increasing the sampling height to $z = [0, 319]$ pixels. The sampling strategy remains unchanged, resulting in an equal convergence time.

For our model training, we use the Adam optimizer with beta of (0.9, 0.999), analogously to ref. 34, and apply gradient clipping of 0.1, to mitigate divergence. We train our model for 100,000 iterations until we reach convergence. We found that prioritizing the boundary condition at the beginning of the training leads to a faster convergence. We set $\lambda_{B0} = 1{,}000$ and exponentially decay it to 1 over the first 50,000 iterations. Similarly, we exponentially decay the learning rate from 5×10^{-4} to 5×10^{-5} over the first 70,000 iterations. The choice of λ_{ff} and λ_{div} is discussed in 'Trade-off between observation and physical model'.

To employ the metrics used for quantifying the quality of our NLFF modelling, we sample the full simulation volume and compute the derivatives using finite differences. From this, we compute $\nabla \cdot \mathbf{B}$ and $\nabla \times \mathbf{B}$.

As an indicator of the degree of force freeness of the obtained solution, the current-weighted average of the angle between the current density and the magnetic field is used[21], where

$$\theta_J = \sin^{-1} \frac{\sum_i J_i \sigma_i}{\sum_i J_i}. \tag{9}$$

and

$$\sigma_i = \frac{|| \mathbf{J} \times \mathbf{B} ||_i}{|| \mathbf{J} ||_i \, || \mathbf{B} ||_i}, \tag{10}$$

and where we employ the corresponding average $\langle \theta_J \rangle$ over all grid points i.

We denote the difference, including the error uncertainty $\mathbf{B}_{\mathrm{error}}$, at each point r within the observed data $\mathbf{B}_0$ as

$$\mathbf{B}_{\mathrm{diff}}(r) = \mathrm{abs}(\mathbf{B}(r) - \mathbf{B}_0(r)) - \mathbf{B}_{\mathrm{error}}(r). \tag{11}$$

For computing $\langle B_{\mathrm{diff}} \rangle$, we first set negative values to 0 and compute the vector norm, such that we estimate the deviation from a tolerance range. Averaging over the full simulation volume then gives $\langle B_{\mathrm{diff}} \rangle$.

The total magnetic energy is computed by integrating over the full simulation volume (V)

$$E = \int_V \frac{B^2}{8\pi} \mathrm{d}V, \tag{12}$$

where we denote E_{FF} when derived from our force-free solution and E_{PF} when derived from the potential-field solution. The free magnetic energy is estimated from the difference between the force-free and potential energy

$$E_{\mathrm{free}} = E_{\mathrm{FF}} - E_{\mathrm{PF}}. \tag{13}$$

For integrated maps of free energy, we apply the Green's function method with the bottom magnetic field of our extrapolation to obtain the potential magnetic field. The potential field for the evaluation of the free-magnetic-energy evolution is obtained by the computationally faster approach based on Fourier expansion in ref. 36.

We note that all metrics can be calculated using the automatic differentiation of the neural representation at each coordinate point, which leads to similar results.

Time-integrated maps

As an observational reference, we use EUV filtergrams from the AIA[22] onboard the SDO. For the time-integrated EUV emission in Fig. 3b, we use observations from flare start to end at 1 min cadence. We use EUV observations from the AIA 94 Å channel, sensitive to the hot flare plasma (peak formation temperature $T = 7$ MK), reproject them to the corresponding SHARP map and normalize for exposure time. From the set of images, we compute the average per pixel, where we mask saturated pixels (>3,000 DN s^{-1}). The images are plotted using a logarithmic normalization over the value range [10, 1,000]. For the maps of magnetic energy release, we use the NLFF solutions temporally closest to the nominal GOES start and end times of the flare. We compute the difference per grid cell, set positive differences to zero and integrate along the individual spatial axis. The energies are normalized linearly between 0 and the maximum value of the map that is obtained from integrating along the z axis (x–y projection).

Magnetic helicity

Given that the boundary-normal components of $\mathbf{B}$ and $\mathbf{B}_{\mathrm{p}}$ match on the surface of the simulation, the gauge-invariant relative magnetic helicity can be written as[37,38]

$$H_{\mathrm{V}} = \int_V (\mathbf{A} + \mathbf{A}_{\mathrm{p}}) \cdot (\mathbf{B} - \mathbf{B}_{\mathrm{p}}) \, \mathrm{d}V, \tag{14}$$

where $\mathbf{A}$ and $\mathbf{A}_{\mathrm{p}}$ are the respective vector potentials satisfying $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{B}_{\mathrm{p}} = \nabla \times \mathbf{A}_{\mathrm{p}}$, as well as $\nabla \cdot \mathbf{A} = 0$ and $\nabla \cdot \mathbf{A}_{\mathrm{p}} = 0$, respectively. H_{V} in equation(14) can be decomposed as $H_{\mathrm{V}} = H_{\mathrm{J}} + H_{\mathrm{PJ}}$ (refs. 39,40), where

$$H_{\mathrm{J}} = \int_V (\mathbf{A} - \mathbf{A}_{\mathrm{p}}) \cdot (\mathbf{B} - \mathbf{B}_{\mathrm{p}}) \, \mathrm{d}V, \tag{15}$$

is the magnetic helicity of the current-carrying field, $\mathbf{B}_{\mathrm{J}} = \mathbf{B} - \mathbf{B}_{\mathrm{p}}$, and

$$H_{\mathrm{PJ}} = 2 \int_V \mathbf{A}_{\mathrm{p}} \cdot (\mathbf{B} - \mathbf{B}_{\mathrm{p}}) \, \mathrm{d}V, \tag{16}$$

is the volume-threading helicity. The vector potentials $\mathbf{A}$ and $\mathbf{A}_{\mathrm{p}}$ are computed using the Coulomb-gauge method in ref. 41.

Series training

A primary aspect of force-free magnetic-field extrapolations is the application to time series. Existing methods perform the simulation from scratch (for example, refs. 24,25). Here we make use of the relatively small changes in time, by iteratively using the weights of the previous time step as initial weights for training. We find that the neural network rapidly adapts to the changes of the photospheric magnetic field, such that we obtain a result comparable to a simulation from scratch (Fig. 3), in a fraction of the time. For the simulation of a time series, we start by training a neural network from scratch for the first magnetogram of the series. The resulting weights are used as starting point for the subsequent extrapolation (analogously to transfer learning), where we set $\lambda_{B0} = 1$ and train with the new boundary magnetic field for 2,000 iterations.

For our experiments, we used four NVIDIA V100 GPUs. The initial simulation is performed in about 45 min. The consecutive simulations

require approximately 1 min per extrapolation (including data pre-processing and calculating the potential-field boundary). This result already allows for operational magnetic-field extrapolations in real time, even with less computing power. A strategy could be to perform the initial extrapolations with the first sight of active regions on the east limb. Afterwards, the proceeding time steps can be computed at a fraction of the instrumental cadence. A limitation is the initial training time, which delays the extrapolation results. The use of better initial weights, for example, by using meta-learning, could be investigated to further speed-up the initial training time.

Analytical field solution

To solve the semi-analytical magnetic fields, we use the same training strategy as for the observational data, where we set $\lambda_{\mathrm{div/ff}} = 0.1$. For C1, the side and top boundaries are provided from the analytical solution. For C2, we use a potential-field boundary.

For the quantitative evaluation, we employ the metrics from ref. 20. We denote the magnetic-field vectors at point i as $\mathbf{B}_i$ and $\mathbf{b}_i$ for the analytical and our simulated magnetic field, respectively. The vector correlation (C_{vec}) compares the local characteristics of the magnetic field

$$C_{\mathrm{vec}} = \frac{\sum_i \mathbf{B}_i \cdot \mathbf{b}_i}{\left(\sum_i \| \mathbf{B}_i \|^2 \sum_i \| \mathbf{b}_i \|^2\right)^{1/2}}. \tag{17}$$

This metric ranges from 0 (parallel vectors) to 1 (identical). The second metric measures the angle between the magnetic-field solutions based on the Cauchy–Schwarz inequality

$$C_{\mathrm{CS}} = \frac{1}{M} \sum_i \frac{\mathbf{B}_i \cdot \mathbf{b}_i}{\| \mathbf{B}_i \| \| \mathbf{b}_i \|} = \frac{1}{M} \sum_i \cos \theta_i, \tag{18}$$

where M refers to the total number of grid points in the volume, and θ_i to the angle between the magnetic-field vectors at point i. The same as for C_{vec}, this metric ranges from 0 to 1. To measure the difference per vector, we compute the mean error normalized by the average vector norm

$$E_{\mathrm{n}} = \frac{\sum_i \| \mathbf{b}_i - \mathbf{B}_i \|}{\sum_i \| \mathbf{B}_i \|}, \tag{19}$$

and the mean error normalized per vector

$$E_{\mathrm{m}} = \frac{1}{M} \sum_i \frac{\| \mathbf{b}_i - \mathbf{B}_i \|}{\| \mathbf{B}_i \|}. \tag{20}$$

Here the best performance is achieved when $E_{\mathrm{m}} = E_{\mathrm{n}} = 0$. We set $E'_{\mathrm{m,n}} = 1 - E_{\mathrm{m,n}}$, such that the best attainable performance corresponds to 1, to allow for an easier comparison with the previous metrics.

In terms of global metrics, we compare the total magnetic-field energy. To this aim, we compute the modelled magnetic energy relative to the input magnetic energy

$$\epsilon = \frac{\sum_i \| \mathbf{b}_i \|^2}{\sum_i \| \mathbf{B}_i \|^2}, \tag{21}$$

where the best performance equals to 1. Values >1 and <1 correspond to an over- or under-estimation, respectively.

In addition, we compute the force freeness of the modelled magnetic field, where we use the current-weighted average sine of the the angle between the magnetic-field and current vector (compare, ref. 20)

$$\sigma_J = \left(\sum_i \frac{\| \mathbf{J}_i \times \mathbf{B}_i \|}{\| \mathbf{B}_i \|} \right) / \sum_i \| \mathbf{J}_i \| . \tag{22}$$

Data availability

All our simulation results are publicly available (parameter variation, time series, 66 individual active regions) at https://doi.org/10.6084/m9.figshare.21983486. See also the project page at https://github.com/RobertJaro/NF2. The SDO HMI and AIA data are provided by JSOC (http://jsoc.stanford.edu/). We provide automatic download scripts with SunPy[42,43].

Code availability

Our codes are publicly available. We provide Python notebooks that perform simulations for arbitrary regions without any pre-requirements. We provide GPU accelerated code for computing the potential-field solution based on the Green's function method[32] at https://github.com/RobertJaro/NF2.

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Author contributions

R.J. developed the method and led the writing of the paper. J.K.T. performed the evaluation and comparison to existing NLFF methods, and contributed to the writing of the paper. A.M.V. contributed to the conceptualization of the study and writing of the paper. T.P. contributed to the HPC computations. All authors discussed the results and commented on the paper.

Competing interests

The authors declare no competing interests.

Additional information

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